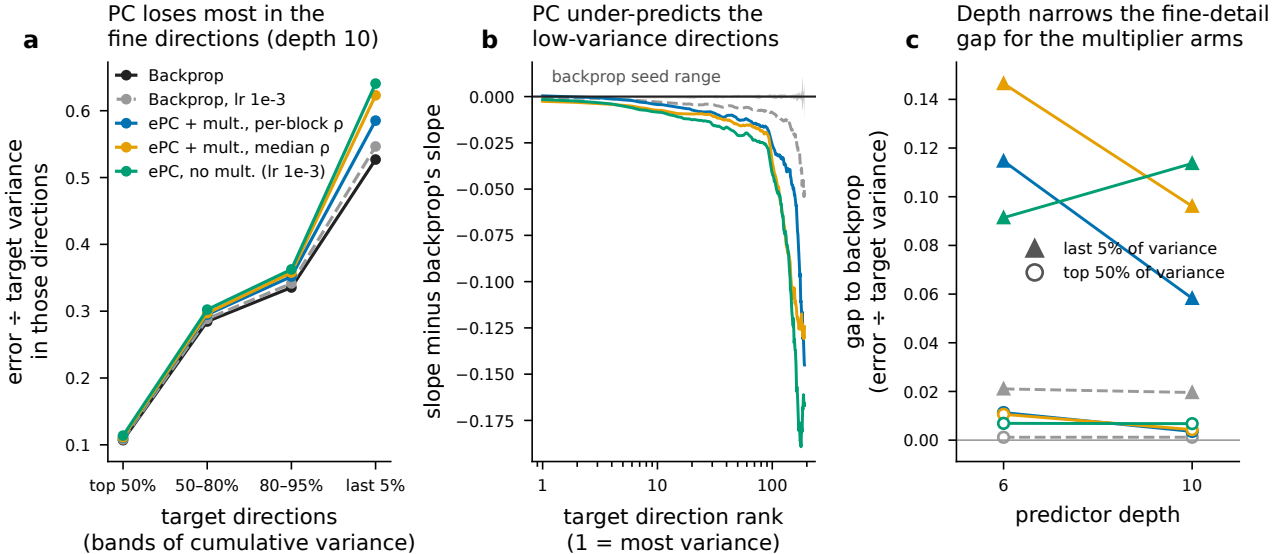
a
PC loses most in the fine directions (depth 10)
Backprop
Backprop, lr 1e-3
ePC + mult., per-block ρ
ePC + mult., median ρ
ePC, no mult. (lr 1e-3)
error ÷ target variance in those directions
0.6
0.5
0.4
0.3
0.2
0.1
top 50%
50–80%
80–95%
last 5%
target directions (bands of cumulative variance)
b
PC under-predicts the low-variance directions
backprop seed range
slope minus backprop's slope
0.000
−0.025
−0.050
−0.075
−0.100
−0.125
−0.150
−0.175
1
10
100
target direction rank (1 = most variance)
c
Depth narrows the fine-detail gap for the multiplier arms
gap to backprop (error ÷ target variance)
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
last 5% of variance
top 50% of variance
6
10
predictor depth